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STAFF REPORT FOR THE 2017 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Tajikistan's risk of external debt distress has increased from low to high in the baseline scenario. The economy's external position remains sensitive to external shocks. Further, there are significant vulnerabilities—mainly arising from the banking sector—that can affect the public debt dynamics. Tajikistan's CPIA-based classification has been recently downgraded to weak performer, reflecting a lower debt capacity.[1] *Medium- and long-term fiscal sustainability can be promoted through adhering to prudent borrowing policies, building fiscal buffers, taking decisive actions to address financial vulnerabilities, and implementing structural reforms to address fiscal risks and unlock growth potential.*

[1] The low-income country debt sustainability framework (LIC DSF) recognizes that better policies and institutions allow countries to manage higher levels of debt, and thus the threshold levels are policy-dependent. Tajikistan's policies and institutions, as measured by the World Bank's Country Policy and Institutional Assessment (CPIA), are classified as a "weak performer", with an average rating of 3.00 during 2014-16. The relevant indicative thresholds for this category are: 27 percent for the PV of PPG external debt to GDP + remittances, 80 percent for the PV of PPG external debt in percent of exports + remittances, 200 percent for the PV of external debt in percent of revenue, 12 percent for the ratio of external debt service to exports plus remittances, and 18 percent for the external debt service to revenue ratio. Also, the benchmark for the PV of total public debt in percent of GDP is 38 percent.

BACKGROUND

1. The current DSA uses two scenarios—a baseline scenario that encompasses current policies, and an alternative, strong-reform, scenario assuming aggressive reforms. It is based on Tajikistan's current CPIA performance classification which is lower than its 2015 classification. The analysis concludes that Tajikistan's risk of debt distress is high (see Figure 1) under the baseline scenario as two debt indicators—the present value (PV) of public and publicly guaranteed (PPG) external debt in percent of the sum of GDP and remittances and the PPG external debt service to revenue—breach the indicative thresholds for several years. Given the importance of remittances in the economy, the calculation of debt indicators incorporates remittances.

2. The public debt stock is projected to increase substantially in the short- to medium terms, driven by large energy sector investment, fiscal costs related to the banking sector, and continued exchange rate depreciation (Figure 2). Over the past five years, Tajikistan's public debt increased from a modest level to 42.3 percent of GDP at end-2016, and PPG external debt at end-2016 increased to 32.9 percent of GDP. Public debt is projected to peak at more than 50 percent of GDP in the medium term under the baseline scenario, and stay slightly lower (49 percent of GDP) in the strong reform scenario. Meanwhile, it is projected that additional public funding of bank recapitalization will be needed. Thus, domestic debt, which stood at 9.4 percent of GDP at end-2016, is projected to increase to 14 percent of GDP in the medium term.

3. Tajikistan remains vulnerable to additional factors. SOEs continue to perform poorly and contingent liabilities have increased. The authorities are also planning to develop a large energy project that would require substantial external borrowing.

MACROECONOMIC ASSUMPTIONS

4. Growth is projected to decline in 2017 and over the medium term, while the fiscal deficit declines and inflation remains generally low. Real GDP growth increased to 6.9 percent in 2016, as higher investment offset the decline in remittances, consumption, and imports. Although investment supported growth in 2016, in the baseline scenario growth is expected to slow somewhat in 2017 and over the medium to long term, initially due to fiscal adjustment and banking sector difficulties and throughout by limited structural reform. Fiscal consolidation will reduce the fiscal deficit in 2017 and stabilize it over the medium term. After an increase in 2017, inflation is projected to decline and remain in single digits. The real effective exchange rate is assumed to remain broadly stable reflecting the external sector assessment that finds no significant exchange rate misalignment.

5. Compared to the 2015 DSA, the current macroeconomic framework projects a less pronounced near-term slowdown in growth but lower growth over the medium and long terms. This reflects two principal factors. First, real GDP growth was higher in 2015 and 2016 than earlier projected,

as higher investment compensated for lower consumption.[2] Over the medium and long terms, the higher-than-previously-expected banking sector losses and limited reforms are projected to lead to lower growth.

6. Pressures in the banking system are likely to affect economic prospects. In 2016, NPLs increased to 54 percent from 30 percent the previous year. The government's decision to recapitalize banks led to exchange rate and inflationary pressures in early 2017. The pressures in the financial system are expected to reduce economic confidence and growth.

7. After a challenging 2015-16, the external position is projected to stabilize over the medium term. Under the baseline scenario, the current account deficit is expected to widen from 3.8 percent of GDP in 2016 to 6.5 percent in 2017, reflecting the liquidity created by the bank recapitalization and imports for a large energy project, and subsequently to decline to around 4.5 percent in the outer years. Exports, which are concentrated on primary goods, would perform modestly despite a boost from expected electricity exports as a large energy project is completed. The dollar value of remittances, which continued to decline in 2016, is projected to rebound over the medium term on the back of economic recovery in the source countries and initiatives to diversify countries receiving Tajik migrant workers. With FDI and debt inflows covering current account deficit, reserves are expected to increase to more than 3.5 months of imports over the medium term under both scenarios. In the strong reform scenario, which envisages more aggressive reforms, the current account deficit would gradually decline to below 4 percent of GDP in the outer years, and financing needs would be lower given more prudent macroeconomic policies. Nevertheless, the balance of payments remains susceptible to external shocks in both scenarios.

8. In comparison to the previous DSA, the public debt-to-GDP ratio in the current DSA is estimated to be higher over the short and medium terms in the baseline. The higher public debt-to-GDP ratio in the current DSA reflects the impact of bank recapitalization on domestic debt, including 6 percent of GDP in 2016, plus an additional projected 4 percent of GDP in 2017, and 3.5 percent of GDP in 2018 in both scenarios. Domestic debt is used for bank recapitalization and repaid.

9. As in the previous DSA, both the baseline and strong reform scenarios assume external borrowing as part of the financing for the large energy project. In the previous DSA, the medium-term fiscal framework included annual expenditure of about 2 percent of GDP for this project. The current DSA assumes that, in addition to budget resources, the authorities would mobilize $1.2 billion non-concessional external borrowing for this energy project during 2017-28, which is estimated to cost $3.9 billion. While the authorities have already raised $500 million in 2017, staff estimate that additional external financing might be needed in 2018-19.

10. The current projections envisage a significant upfront fiscal adjustment to reduce the fiscal deficit and lower public debt. The baseline macroeconomic framework projects fiscal adjustment of 2 percent of GDP in 2017, net of the cost of bank recapitalization. This adjustment is expected to be achieved mainly through lower capital expenditures. The overall fiscal deficit is projected to remain stable at

[2] The staff team has reviewed the authorities' new GDP estimates using the production and expenditure methods and discussed with STA. Considering this review, staff has revised its estimate of 2015 growth (from the previous Article IV report) upward to equal the authorities' estimates.

below 2 percent of GDP over the medium term and improve over the long term in the baseline scenario, while in the strong reform scenario an additional adjustment of about 1 percent of GDP is projected from 2020.

Tajikistan: Baseline External DSA Compared to Previous DSA, 2015–34 1/
(In percent of GDP, unless otherwise indicated)

	DSA May 2015		Current DSA		(current vs. previous)	
	2015-19	2020-34	2015-19	2020-34	2015-19	2020-34
Real GDP growth, percent	4.4	5.0	5.1	4.0	0.7	-1.0
GDP deflator, percent	7.9	6.0	5.5	6.0	-2.4	0.0
Nominal GDP (billions somonis)	67.3	216.5	62.1	181.1	-5.2	-35.4
Revenue and grants	26.3	26.2	28.9	30.3	2.6	4.1
Expenditure and net lending	28.6	29.6	34.2	32.7	5.6	3.1
Overall balance (incl. PIP)	-2.3	-3.5	-5.3	-2.3	-3.0	1.2
Public debt	30.4	33.0	43.6	40.9	13.2	7.9
Domestic debt	3.2	2.8	10.2	3.9	7.0	1.1
External debt	48.0	49.5	38.2	42.7	-9.8	-6.8
of which: Large Development Projects			2.3	0.4	2.3	0.4
Exports of goods and services	19.9	25.0	12.5	11.3	-7.4	-13.7
Imports of goods and services	-48.4	-51.0	-47.3	-52.2	1.1	-1.2
Current account balance	-5.1	-2.9	-5.6	-4.4	-0.5	-1.5

Sources: Authorities, IMF, and World Bank staff.
[1] The current DSA used new BOP data for 2014-16 based on STA TA.

EXTERNAL DSA

Baseline Scenario

11. Based on Tajikistan's current CPIA-based performance classification, two debt indicators breach indicative thresholds under the baseline, while all debt indicators would breach the thresholds under stress tests. The PV of PPG external debt-to-(GDP + remittances) ratio is projected to peak at 28.1 percent in 2019 and stay above or upon the indicative threshold (27 percent) for four years before declining gradually to below 20 percent in the outer years. Meanwhile, the PV of external debt-to-exports + remittances ratio is projected to peak at 77 percent in 2019, and the PV of external debt-to-revenue would peak at 151 percent, but both indicators are below the respective thresholds. Debt service indicators would be significantly higher than average in 2025-27, because of large repayments in those years. The external debt- service-to-exports + remittances ratio would peak at 11 percent in 2025, and debt service-to-revenue ratio would breach the threshold in 2025 at 20.6 percent and remain near this level through 2027. In the most extreme stress tests, the PV of external debt to GDP + remittances, the PV of external debt-to-exports + remittances, the PV of external debt-to-revenue, external debt-service-to-exports + remittances and external debt-service-to-revenue all breach the indicative thresholds with large margins with the most severe shocks.

Tajikistan: Key Macroeconomic Data for Debt Sustainability Analysis Scenario 1/
(In percent of GDP)

	2015	2016	2019	2024	2027	2037
	Est.	Est.	Proj.	Proj.	Proj.	Proj.
			Baseline			
Real GDP growth, percent	6.0	6.9	4.0	4.0	4.0	4.0
GDP deflator, percent	0.1	5.3	6.0	6.0	6.0	6.0
Revenue and grants	29.9	28.8	29.4	29.3	31.9	30.8
Expenditure and net lending	31.8	39.4	31.3	31.5	35.7	32.1
Overall balance (incl. PIP and stat. discrepancy) [2]	-1.9	-10.6	-1.9	-2.2	-3.8	-1.3
Exports of goods and services	10.5	13.3	12.6	12.1	11.3	9.3
Imports of goods and services	42.2	43.0	53.5	52.7	51.7	50.3
Current account balance	-6.0	-3.8	-5.6	-4.2	-4.0	-4.5
Gross international reserves	2.0	2.3	2.8	3.3	4.4	4.4
			Strong Reform			
Real GDP growth, percent	6.0	6.9	5.5	6.0	6.0	6.0
GDP deflator, percent	0.1	5.3	6.0	6.0	6.0	6.0
Revenue and grants	29.9	27.9	29.4	28.7	32.6	31.4
Expenditure and net lending	31.8	38.5	30.8	30.9	34.7	31.2
Overall balance (incl. PIP and stat. discrepancy) [2]	-1.9	-10.6	-1.4	-2.2	-2.1	0.3
Exports of goods and services	10.5	13.3	12.3	10.7	9.4	6.5
Imports of goods and services	42.2	43.0	51.1	46.6	43.8	35.2
Current account balance	-6.0	-3.8	-4.3	-3.6	-3.7	-3.4
Gross international reserves	2.0	2.3	2.8	3.6	4.5	4.4

Sources: Authorities, IMF, and World Bank staff.

[1] The current DSA used new BOP data for 2014-16 based on STA TA.

[2] Overall balance in 2016 includes 6.1 percent of GDP of bank recapitalization in addition to 4.5 percent of GDP deficit from regular fiscal operations.

12. Tajikistan's external position continues to be susceptible to negative shocks. The increased reliance on external financing and vulnerability to remittances shocks and exchange rate depreciation suggest the need to broaden the revenue base, diversify exports, and deepen reforms to improve the policy framework and institutional capacity. Further, to assure debt sustainability, it is also important that the authorities limit contracting debt on non-concessional terms in the future.

Strong-Reform Scenario

13. Debt indicators stay below the thresholds in the strong reform scenario, but under the stress test most debt indicators breach thresholds. The PV of PPG external debt-to-GDP + remittances ratio is projected to decline gradually from a peak of 27 percent in 2019 to 14 percent by end–2037. The PV of external debt-to-exports + remittances ratio is expected to decline monotonically from 77 percent to 53 percent by the end of the projection period, while the PV of external debt-to-revenue will drop from 140 percent to below 60 percent. Like the baseline scenario, external debt service indicators would spike in 2025–27 but remain below the thresholds. Under the stress test, PV of external debt to GDP + remittances,

PV of external debt to exports + remittances and external debt service-to-revenue would breach the indicative thresholds with the most severe shocks, while external debt service-to-exports + remittances and PV of external debt-to-revenue would breach the thresholds by only small margins at their peaks.

PUBLIC DSA

14. Public debt indicators are projected to increase over the medium term, reflecting financing for the large energy project and additional domestic debt of about 7.5 percent of GDP in 2017-18 for bank recapitalization. Under the baseline scenario, the PV of public debt would rise from 35.2 percent of GDP in 2016 to a peak of 50.7 percent in 2019 and is projected to be above the benchmark (38 percent for a weak policy performer) for nine years until 2026. Under the strong-reform scenario, the PV of public debt would peak at 48.8 percent of GDP in 2018, and stay above the benchmark until 2022. In addition, the liabilities of SOEs present a fiscal risk, which if realized would increase the public debt level.

15. Standard stress tests indicate that public debt is sensitive to potential shocks. As a large proportion of public debt is denominated in foreign currency, exchange rate changes can adversely affect the government's debt level. The most extreme stress test—which is a 30 percent depreciation of the nominal exchange rate—shows the PV of the debt-to-GDP ratio to peak at 65 percent and remain above the benchmark across the whole projection period.

FISCAL CONTINGENT LIABILITIES

16. There are substantial fiscal contingent liabilities arising from the poorly managed SOE and banking sectors. Contingent liabilities, equivalent to 8.5 percent of GDP in 2016, could be added to public debt should the associated risks materialize. This figure is based on information for the stock of accrued liabilities of the 23 largest SOEs at end-December 2016 plus all deposits in Amonat Bank, which are government guaranteed as they are not insured by the deposit insurance fund. These liabilities could add to the already high risk of debt distress.

CONCLUSION

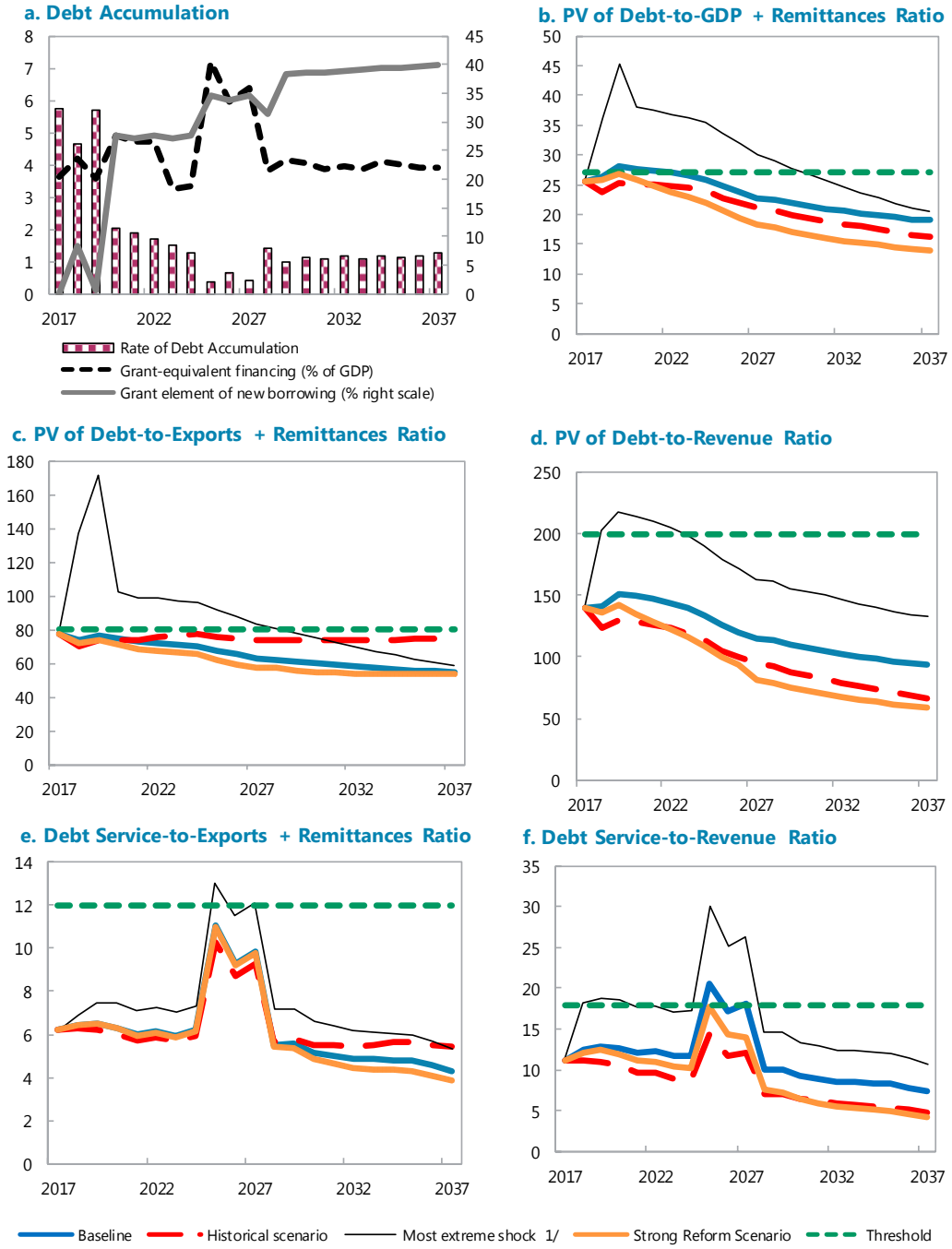
17. The DSA, which uses the thresholds of a CPIA performance rating of a weak policy performer, concludes that Tajikistan's risk of debt distress is high. Under the baseline scenario, two debt indicators breach the indicative thresholds, and all debt indicators breach the thresholds under the stress test. Public debt indicators are expected to increase over the medium term given the projected costs of bank recapitalization and external financing for a large energy project and remain above the benchmark for a few years, while the most severe shock would cause public debt to stay above the benchmark across the projection period under the baseline scenario.

18. Tajikistan's risk of debt distress would be lower in a strong-reform scenario. Under the strong-reform scenario, baseline indicators stay below the thresholds, but stress tests of debt indicators still show a breach of the thresholds.

19. Considering the risks, the government should develop a prudent macro-fiscal framework to contain debt risks. The authorities will need to reduce the fiscal deficit, avoid contracting and guaranteeing large amounts of external debt on non-concessional terms, address the vulnerabilities in the financial sector and quasi-fiscal liabilities arising from SOEs, and allow a margin to absorb the debt impact of potential economic shocks. Also, the government should seek to broaden the revenue and export bases through diversification and continued reforms to improve the policy framework and institutional capacity, which would lead to a higher growth path over the medium and long run. Further, the authorities should augment their debt management capacity. It is important that these issues be addressed and that contingent liabilities be managed actively so they do not add to the state's liabilities.

20. The authorities' views. The authorities consider that the debt risk remains moderate and are committed to keeping external debt below 40 percent of GDP over the medium term, the upper limit set under Tajikistan's debt management strategy. Regarding domestic debt, the authorities disagree with the assumption of public sector led bank recapitalizations in 2017–18 because of the harmful macroeconomic effects of the one carried out in 2016. They also maintain that the state will not take on SOEs liabilities because SOEs are solely responsible for their liabilities under Tajikistan laws. Therefore, they expect domestic debt to begin declining in 2017 as the recapitalized banks repay the outstanding bonds and expect no significant fiscal risks emerging from SOEs liabilities. They also emphasize that staff's macro projections might be too conservative—they expect annual real GDP growth at or above 7 percent per year over the medium and long terms, as opposed to staff's baseline projection of 4 percent, and consider that remittances growth is likely to be stronger than the staff projection. In the meantime, the authorities are actively seeking external financing, both on concessional and non-concessional terms, for the large energy project. The authorities believe that there is fiscal space for additional external borrowing for this project based on the macroeconomic indicators and their projections. They note that the attracted external resources are primarily used in strategic infrastructure projects financing and in implementation of reforms, which in turn will stimulate economic growth in the long term and strengthen the economy.

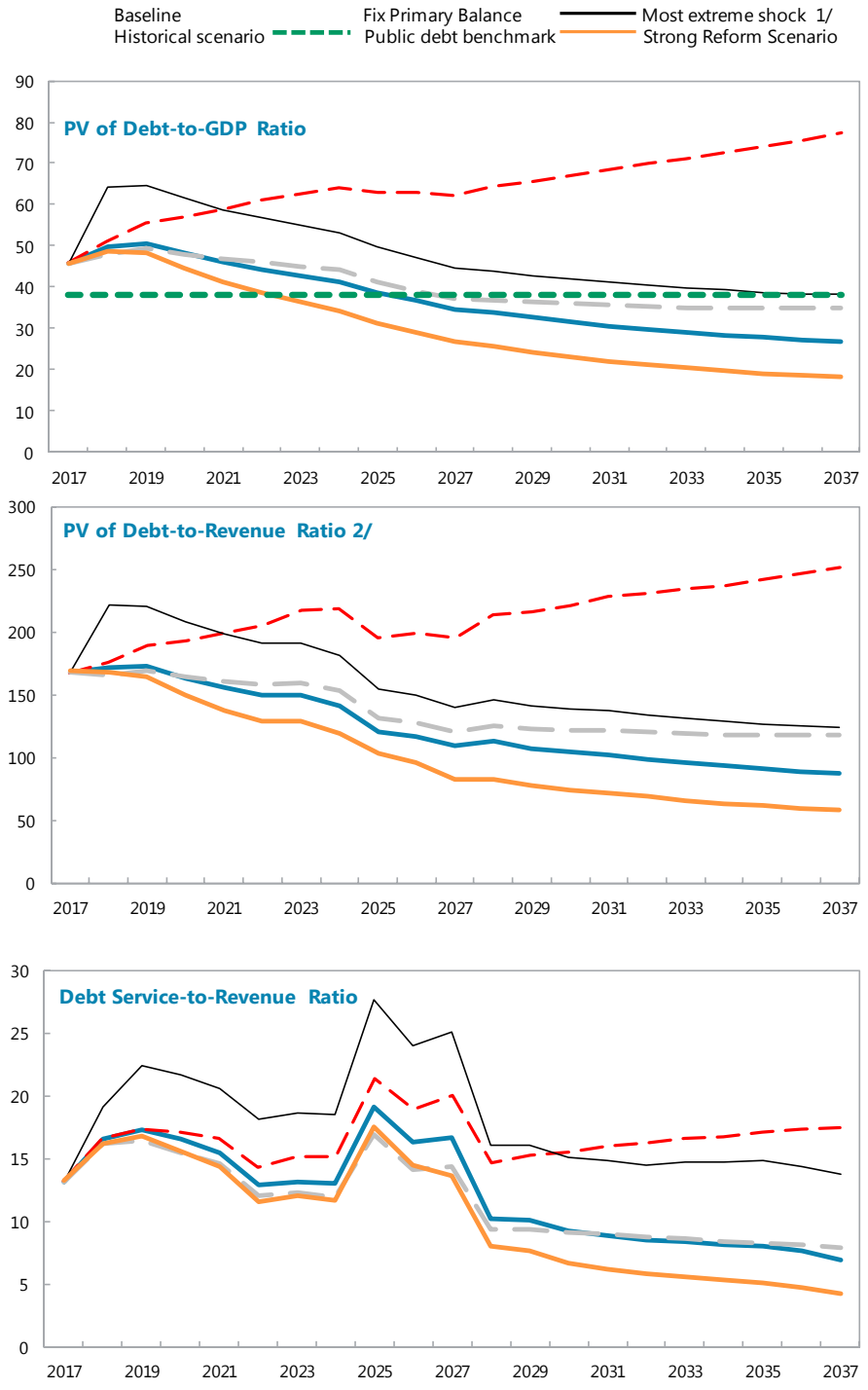
Figure 1. Tajikistan: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2017–37 1/ -Baseline



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027. In figure b. it corresponds to a Non-debt flows shock; in c. to a Non-debt flows shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Tajikistan Indicators of Public Debt Under Alternative Scenarios, 2017–37 1/ -Baseline



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.

Table 1. Tajikistan: External Debt Sustainability Framework, Baseline Scenario, 2014–37 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections								
	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2027	2037
External debt (nominal) 1/	**44.7**	**52.3**	**57.8**			**62.2**	**63.1**	**61.6**	**60.0**	**57.9**	**55.7**		**48.0**	**40.2**
of which: public and publicly guaranteed (PPG)	24.4	31.5	32.9			39.7	42.1	44.6	45.0	45.1	45.0		43.1	38.0
Change in external debt	-2.3	7.6	5.5			4.3	0.9	-1.4	-1.7	-2.0	-2.2		-1.8	-0.2
Identified net debt-creating flows	-4.3	8.6	6.0			0.8	0.5	0.1	-0.4	-0.8	-1.0		-0.8	0.6
Non-interest current account deficit	**1.4**	**3.6**	**2.6**	**4.9**	**4.2**	**5.1**	**4.8**	**3.3**	**2.5**	**2.3**	**2.2**		**2.5**	**2.3**
Deficit in balance of goods and services	35.8	31.8	29.6			33.4	38.7	41.0	41.7	41.2	40.8		40.5	41.0
Exports	9.0	10.5	13.3			12.8	13.1	12.6	12.2	13.1	12.7		11.3	9.3
Imports	44.8	42.2	43.0			46.2	51.7	53.5	53.9	54.2	53.5		51.7	50.3
Net current transfers (negative = inflow)	-15.7	-10.7	-11.3	-25.9	17.7	-14.2	-16.2	-16.9	-17.2	-17.1	-16.7		-15.4	-13.6
of which: official	-0.6	-0.5	-0.6			-0.6	-0.6	-0.6	-0.6	-0.5	-0.5		-0.4	-0.3
Other current account flows (negative = net inflow)	-18.7	-17.4	-15.8			-14.1	-17.7	-20.8	-22.0	-21.8	-21.9		-22.6	-25.0
Net FDI (negative = inflow)	**-3.3**	**-5.4**	**-5.0**	**-4.0**	**2.9**	**-3.0**	**-3.2**	**-3.2**	**-3.2**	**-3.2**	**-3.2**		**-3.0**	**-2.4**
Endogenous debt dynamics 2/	**-2.3**	**10.4**	**8.4**			**-1.2**	**-1.0**	**0.0**	**0.3**	**0.2**	**0.0**		**-0.3**	**0.7**
Contribution from nominal interest rate	1.4	2.5	1.3			1.3	1.5	2.4	2.6	2.4	2.2		1.5	2.2
Contribution from real GDP growth	-2.9	-3.2	-4.1			-2.5	-2.5	-2.4	-2.3	-2.3	-2.2		-1.9	-1.5
Contribution from price and exchange rate changes	-0.8	11.0	11.2			...	...	...	...	-1.3	...		...	...
Residual (3-4) 3/	**2.0**	**-0.9**	**-0.5**			**3.5**	**0.4**	**-1.5**	**-1.3**	0.1	**-1.2**		**-0.9**	**-0.8**
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
PV of external debt 4/	...	...	50.8			55.6	56.8	56.0	53.6	51.1	48.3		36.4	28.6
In percent of exports	...	...	380.7			433.1	434.7	444.8	439.7	390.0	380.5		323.5	307.0
PV of PPG external debt	**...**	**...**	**25.8**			**33.1**	**35.8**	**39.0**	**38.7**	**38.2**	**37.6**		**31.6**	**26.4**
In percent of exports	**...**	**...**	**193.6**			**258.0**	**274.3**	**309.8**	**317.2**	**292.0**	**296.0**		**280.6**	**282.9**
In percent of government revenues	**...**	**...**	**100.0**			**139.1**	**140.8**	**151.0**	**149.3**	**146.6**	**143.5**		**114.6**	**93.4**
Debt service-to-exports ratio (in percent)	**26.9**	**34.6**	**20.8**			**43.8**	**44.2**	**50.0**	**48.5**	**41.5**	**39.7**		**54.1**	**42.7**
PPG debt service-to-exports ratio (in percent)	**15.2**	**15.4**	**15.3**			**20.7**	**24.2**	**26.4**	**27.0**	**24.0**	**25.2**		**44.3**	**22.3**
PPG debt service-to-revenue ratio (in percent)	**5.1**	**6.0**	**7.9**			**11.2**	**12.4**	**12.9**	**12.7**	**12.1**	**12.2**		**18.1**	**7.4**
Total gross financing need (Billions of U.S. dollars)	0.0	0.1	0.0			0.6	0.5	0.5	0.4	0.4	0.4		0.7	0.9
Non-interest current account deficit that stabilizes debt ratio	3.7	-4.0	-2.9			0.7	3.9	4.7	4.2	4.3	4.4		4.2	2.5
Key macroeconomic assumptions														
Real GDP growth (in percent)	6.7	6.0	6.9	6.8	1.2	4.5	4.0	4.0	4.0	4.0	4.0	4.1	4.0	4.0
GDP deflator in US dollar terms (change in percent)	1.8	-19.8	-17.6	3.5	15.3	0.0	-5.2	2.5	2.3	2.3	2.2	0.7	1.9	1.9
Effective interest rate (percent) 5/	3.2	4.8	2.2	3.4	1.2	2.3	2.3	4.0	4.4	4.3	4.0	3.6	3.3	5.8
Growth of exports of G&S (US dollar terms, in percent)	-10.7	-1.4	12.0	-3.4	21.4	0.5	0.4	2.8	3.1	14.2	3.1	4.0	3.7	4.7
Growth of imports of G&S (US dollar terms, in percent)	-19.1	-19.9	-10.4	5.1	25.7	12.4	10.5	10.4	7.2	7.0	4.7	8.7	4.5	6.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	0.1	8.4	0.5	27.5	27.1	27.5	15.2	34.5	39.8
Government revenues (excluding grants, in percent of GDP)	26.9	27.0	25.8			23.8	25.4	25.8	25.9	26.1	26.2		27.6	28.3
Aid flows (in Billions of US dollars) 7/	0.1	0.2	0.2			0.4	0.4	0.3	0.4	0.4	0.5		1.0	1.1
of which: Grants	0.1	0.2	0.2			0.3	0.3	0.3	0.3	0.3	0.3		0.5	0.6
of which: Concessional loans	0.0	0.0	0.0			0.1	0.1	0.1	0.2	0.1	0.1		0.5	0.5
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.6	4.2	3.6	4.9	4.7	4.7		6.4	3.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			35.0	40.0	33.9	59.1	60.1	60.9		62.4	65.5
Memorandum items:														
Nominal GDP (Billions of US dollars)	9.2	7.9	6.9			7.2	7.1	7.6	8.1	8.6	9.1		12.2	21.9
Nominal dollar GDP growth	8.7	-15.0	-11.9			4.5	-1.4	6.6	6.4	6.4	6.2	4.8	6.0	6.0
PV of PPG external debt (in Billions of US dollars)			1.8			2.2	2.5	2.9	3.1	3.2	3.4		3.8	5.7
(PVt-PVt-1)/GDPt-1 (in percent)						5.7	4.6	5.7	2.0	1.9	1.7	3.6	0.4	1.3
Gross workers' remittances (Billions of US dollars)	3.4	2.3	1.9			2.2	2.6	2.9	3.2	3.4	3.6		4.8	8.5
PV of PPG external debt (in percent of GDP + remittances)	...	...	20.3			25.4	26.4	28.1	27.7	27.4	27.0	27.0	22.7	19.0
PV of PPG external debt (in percent of exports + remittances)	...	...	64.0			77.0	73.3	76.3	74.4	72.5	72.0		62.5	54.7
Debt service of PPG external debt (in percent of exports + remittances)	...	...	5.1			6.2	6.5	6.5	6.3	6.0	6.1		9.9	4.3

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Tajikistan: Public Sector Debt Sustainability Framework, Baseline Scenario, 2014–37

(In percent of GDP, unless otherwise indicated)

	Actual			Average 5/	Standard Deviation 5/	Estimate		Projections				2017-22 Average		
	2014	2015	2016			2017	2018	2019	2020	2021	2022		2027	2037
Public sector debt 1/	27.9	34.7	42.3			52.4	56.2	56.3	54.6	52.9	51.8		46.3	38.5
of which: foreign-currency denominated	24.4	31.5	32.9			39.7	42.1	44.6	45.0	45.1	45.0		43.1	38.0
Change in public sector debt	-1.1	6.8	7.6			10.2	3.8	0.1	-1.7	-1.7	-1.1		-1.1	-0.4
Identified debt-creating flows	-1.0	7.6	10.3			11.0	5.3	-2.0	-1.8	-1.8	-1.7		1.0	-0.9
Primary deficit	-0.4	1.3	10.0	2.9	3.4	5.5	4.0	0.2	0.0	-0.1	-0.1	1.6	2.7	0.8
Revenue and grants	28.4	29.9	28.8			27.4	29.1	29.4	29.6	29.6	29.8		31.9	30.8
of which: grants	1.5	3.0	3.0			3.6	3.6	3.6	3.6	3.6	3.6		4.4	2.5
Primary (noninterest) expenditure	27.9	31.3	38.8			32.8	33.0	29.6	29.5	29.6	29.7		34.6	31.6
Automatic debt dynamics	-0.2	6.3	0.3			1.5	-2.2	-2.2	-1.8	-1.7	-1.6		-1.7	-1.7
Contribution from interest rate/growth differential	-1.9	-1.3	-2.2			-2.2	-1.9	-2.0	-1.8	-1.6	-1.5		-1.7	-1.7
of which: contribution from average real interest rate	-0.1	0.2	0.0			-0.3	0.1	0.2	0.4	0.5	0.5		0.1	-0.2
of which: contribution from real GDP growth	-1.8	-1.6	-2.2			-1.8	-2.0	-2.2	-2.2	-2.1	-2.0		-1.8	-1.5
Contribution from real exchange rate depreciation	1.7	7.6	2.5			3.7	-0.3	-0.2	0.0	-0.1	-0.1		...	...
Other identified debt-creating flows	-0.3	0.0	0.0			4.0	3.5	0.0	0.0	0.0	0.0		0.0	0.0
Privatization receipts (negative)	-0.3	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			4.0	3.5	0.0	0.0	0.0	0.0		0.0	0.0
Residual, including asset changes	-0.1	-0.8	-2.7			-0.8	-1.5	2.1	0.1	0.1	0.5		-2.1	0.5
Other Sustainability Indicators														
PV of public sector debt	...	...	35.2			45.8	49.9	50.7	48.3	46.0	44.4	47.5	34.7	26.9
of which: foreign-currency denominated	...	...	25.8			33.1	35.8	39.0	38.7	38.2	37.6		31.6	26.4
of which: external	...	...	25.8			33.1	35.8	39.0	38.7	38.2	37.6		31.6	26.4
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...
Gross financing need 2/	1.7	3.9	12.5			9.2	8.8	5.3	4.8	4.5	3.7		8.0	3.0
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	122.3			167.3	171.7	172.5	163.4	155.3	149.2		108.8	87.2
PV of public sector debt-to-revenue ratio (in percent)	...	...	136.4			192.6	196.2	196.2	186.3	176.5	169.4		126.0	95.1
of which: external 3/	...	...	100.0			139.1	140.8	151.0	149.3	146.6	143.5		114.6	93.4
Debt service-to-revenue and grants ratio (in percent) 4/	7.3	8.2	8.5			13.2	16.6	17.3	16.5	15.5	12.8		16.7	7.0
Debt service-to-revenue ratio (in percent) 4/	7.8	9.1	9.5			15.2	19.0	19.7	18.8	17.6	14.6		19.4	7.6
Primary deficit that stabilizes the debt-to-GDP ratio	0.6	-5.5	2.5			-4.7	0.2	0.1	1.7	1.6	1.1		3.8	1.2
Key macroeconomic and fiscal assumptions														
Real GDP growth (in percent)	6.7	6.0	6.9	6.8	1.2	4.5	4.0	4.0	4.0	4.0	4.0	4.1	4.0	4.0
Average nominal interest rate on forex debt (in percent)	1.5	1.5	1.5	1.5	0.1	2.2	2.9	3.5	3.7	3.7	3.6	3.2	2.4	1.3
Average real interest rate on domestic debt (in percent)	-1.5	4.3	-1.6	-1.3	3.2	-5.0	-2.4	-2.9	-2.8	-2.6	-2.4	-3.0	-2.0	0.5
Real exchange rate depreciation (in percent, + indicates depreciation)	7.3	32.9	8.4	-0.1	16.1	11.7	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	5.5	0.1	5.3	12.1	9.4	10.0	6.0	6.0	6.0	6.0	6.0	6.7	6.0	6.0
Growth of real primary spending (deflated by GDP deflator, in percent)	9.3	18.6	32.6	6.1	11.2	-11.5	4.6	-6.9	3.8	4.1	4.5	-0.2	6.2	3.9
Grant element of new external borrowing (in percent)	...	...	...	...	...	0.1	8.4	0.5	27.5	27.1	27.5	15.2	34.5	39.8

Sources: Country authorities; and staff estimates and projections.

1/ Coverage of public sector is general government and gross debt is used.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 3. Tajikistan: Sensitivity Analysis for Key Indicators of Public Debt—Baseline, 2017–37

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	46	50	51	48	46	44	35	27
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	46	48	49	48	47	46	37	35
A2. Primary balance is unchanged from 2017	46	51	56	57	59	61	62	77
A3. Permanently lower GDP growth 1/	46	50	51	49	47	46	38	38
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2018-2019	46	49	48	45	43	41	28	16
B2. Primary balance is at historical average minus one standard deviation in 2018-2019	46	52	57	54	52	50	40	31
B3. Combination of B1-B2 using one half standard deviation shocks	46	49	52	49	46	44	30	16
B4. One-time 30 percent real depreciation in 2018	46	64	65	62	59	57	45	38
B5. 10 percent of GDP increase in other debt-creating flows in 2018	46	56	57	55	53	51	41	32
PV of Debt-to-Revenue Ratio 2/								
Baseline	167	172	172	163	155	149	109	87
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	167	165	169	164	160	158	120	118
A2. Primary balance is unchanged from 2017	167	175	189	193	198	205	195	251
A3. Permanently lower GDP growth 1/	167	172	174	165	158	153	119	122
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2018-2019	167	168	165	154	145	137	90	53
B2. Primary balance is at historical average minus one standard deviation in 2018-2019	167	177	193	184	175	168	125	99
B3. Combination of B1-B2 using one half standard deviation shocks	167	170	179	167	156	148	96	51
B4. One-time 30 percent real depreciation in 2018	167	222	220	208	198	191	140	124
B5. 10 percent of GDP increase in other debt-creating flows in 2018	167	194	195	186	178	172	130	104
Debt Service-to-Revenue Ratio 2/								
Baseline	13	17	17	17	16	13	17	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	13	16	16	16	15	12	14	8
A2. Primary balance is unchanged from 2017	13	17	17	17	17	14	20	17
A3. Permanently lower GDP growth 1/	13	17	17	17	16	13	17	9
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2018-2019	13	16	17	16	15	12	16	5
B2. Primary balance is at historical average minus one standard deviation in 2018-2019	13	17	17	17	16	13	18	8
B3. Combination of B1-B2 using one half standard deviation shocks	13	16	17	16	15	13	17	5
B4. One-time 30 percent real depreciation in 2018	13	19	22	22	21	18	25	14
B5. 10 percent of GDP increase in other debt-creating flows in 2018	13	17	18	18	16	14	19	9

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

Table 4. Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt—Baseline, 2017–37

(In percent)

	Projections							
	2017	2018	2019	2020	2021	2022	**2027**	2037
PV of debt-to GDP ratio								
Baseline	33	36	39	39	38	38	32	26
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	33	31	34	33	33	32	26	19
A2. New public sector loans on less favorable terms in 2017-2037 2/	33	37	42	42	43	43	43	43
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	33	35	37	37	36	36	30	25
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	33	38	45	44	44	43	36	27
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	33	38	48	47	47	46	38	32
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	33	43	54	53	52	51	42	28
B5. Combination of B1-B4 using one-half standard deviation shocks	33	35	43	43	42	41	34	27
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	33	52	56	55	55	54	45	37
PV of debt-to-exports ratio								
Baseline	258	274	310	317	292	296	281	283
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	258	240	267	272	251	254	233	201
A2. New public sector loans on less favorable terms in 2017-2037 2/	258	283	331	347	327	340	385	462
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	258	270	305	311	286	290	273	274
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	258	385	652	665	610	618	577	524
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	258	270	305	311	286	290	273	274
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	258	328	427	435	399	404	373	305
B5. Combination of B1-B4 using one-half standard deviation shocks	258	322	471	480	441	447	419	395
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	258	270	305	311	286	290	273	274
PV of debt-to-revenue ratio								
Baseline	139	141	151	149	147	144	115	93
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	139	123	130	128	126	123	95	66
A2. New public sector loans on less favorable terms in 2017-2037 2/	139	145	161	163	164	165	157	153
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	139	136	144	142	139	136	108	88
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	139	148	174	171	168	164	129	95
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	139	149	186	183	179	176	140	113
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	139	168	208	205	200	196	152	101
B5. Combination of B1-B4 using one-half standard deviation shocks	139	137	167	164	161	157	124	95
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	139	203	217	214	210	205	163	132

Table 4. Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt—Baseline, 2017–37 (Concluded)

(In percent)

	Projections							
	2017	2018	2019	2020	2021	2022	**2027**	2037
Debt service-to-exports ratio								
Baseline	21	24	26	27	24	25	44	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	21	22	23	22	19	20	29	15
A2. New public sector loans on less favorable terms in 2017-2037 2/	21	24	23	25	23	24	36	36
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	21	24	26	27	24	25	44	22
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	21	32	49	52	46	48	89	46
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	21	24	26	27	24	25	44	22
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	21	24	28	31	27	28	55	29
B5. Combination of B1-B4 using one-half standard deviation shocks	21	28	37	39	35	36	66	34
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	21	24	26	27	24	25	44	22
Debt service-to-revenue ratio								
Baseline	11	12	13	13	12	12	**18**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	11	11	11	11	10	10	12	5
A2. New public sector loans on less favorable terms in 2017-2037 2/	11	12	11	12	12	12	15	12
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	11	12	12	12	12	12	17	7
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	11	12	13	13	13	13	20	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	11	13	16	16	15	15	23	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	11	12	14	14	14	14	22	10
B5. Combination of B1-B4 using one-half standard deviation shocks	11	12	13	13	13	13	19	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	11	18	19	19	18	18	26	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	25	25	25	25	25	25	25	25

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 5. Tajikistan: External Debt Sustainability Framework, Strong Reform Scenario, 2014–37 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections								
	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2027	2037
External debt (nominal) 1/	**44.7**	**52.3**	**57.8**			**62.2**	**61.4**	**58.0**	**54.1**	**50.4**	**46.8**		**36.0**	**28.4**
of which: public and publicly guaranteed (PPG)	24.4	31.5	32.9			39.7	40.6	41.7	40.4	39.1	37.7		33.1	25.9
Change in external debt	-2.3	7.6	5.5			4.3	-0.7	-3.5	-3.9	-3.7	-3.6		-1.5	-0.4
Identified net debt-creating flows	-4.3	8.6	6.0			0.8	-1.1	-2.0	-2.7	-2.5	-2.5		-0.8	0.2
Non-interest current account deficit	**1.4**	**3.6**	**2.6**	**4.9**	**4.2**	**5.1**	**3.8**	**2.0**	**1.2**	**1.2**	**1.3**		**2.5**	**2.0**
Deficit in balance of goods and services	35.8	31.8	29.6			33.4	37.3	38.8	38.7	37.7	36.8		34.4	28.7
Exports	9.0	10.5	13.3			12.8	12.9	12.3	11.7	12.3	11.7		9.4	6.5
Imports	44.8	42.2	43.0			46.2	50.2	51.1	50.4	50.0	48.5		43.8	35.2
Net current transfers (negative = inflow)	-15.7	-10.7	-11.3	-25.9	17.7	-14.2	-16.0	-16.5	-16.5	-16.0	-15.4		-12.9	-9.4
of which: official	-0.6	-0.5	-0.6			-0.6	-0.6	-0.6	-0.5	-0.5	-0.5		-0.4	-0.2
Other current account flows (negative = net inflow)	-18.7	-17.4	-15.8			-14.1	-17.5	-20.3	-21.0	-20.5	-20.1		-18.9	-17.3
Net FDI (negative = inflow)	**-3.3**	**-5.4**	**-5.0**	**-4.0**	**2.9**	**-3.0**	**-3.2**	**-3.1**	**-3.1**	**-3.0**	**-2.9**		**-2.5**	**-1.7**
Endogenous debt dynamics 2/	**-2.3**	**10.4**	**8.4**			**-1.2**	**-1.7**	**-0.8**	**-0.8**	**-0.8**	**-0.8**		**-0.9**	**-0.1**
Contribution from nominal interest rate	1.4	2.5	1.3			1.3	1.5	2.3	2.4	2.2	2.0		1.2	1.5
Contribution from real GDP growth	-2.9	-3.2	-4.1			-2.5	-3.1	-3.1	-3.2	-3.0	-2.8		-2.1	-1.6
Contribution from price and exchange rate changes	-0.8	11.0	11.2			...	...	...	...	-1.2	...		...	...
Residual (3-4) 3/	**2.0**	**-0.9**	**-0.5**			**3.5**	**0.4**	**-1.5**	**-1.2**	0.0	**-1.1**		**-0.7**	**-0.5**
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
PV of external debt 4/	...	...	50.8			55.6	55.6	53.3	49.1	45.3	41.5		27.1	20.4
In percent of exports	...	...	380.7			433.1	430.3	433.4	420.6	367.8	354.2		287.6	315.6
PV of PPG external debt	**...**	**...**	**25.8**			**33.1**	**34.8**	**37.0**	**35.4**	**33.9**	**32.4**		**24.3**	**17.9**
In percent of exports	**...**	**...**	**193.6**			**258.0**	**269.4**	**300.8**	**303.3**	**275.7**	**276.4**		**257.8**	**277.1**
In percent of government revenues	**...**	**...**	**101.5**			**139.5**	**135.8**	**141.9**	**134.7**	**128.6**	**122.4**		**81.6**	**58.9**
Debt service-to-exports ratio (in percent)	**26.9**	**34.6**	**20.8**			**43.8**	**44.2**	**49.9**	**48.0**	**40.6**	**38.7**		**51.9**	**41.0**
PPG debt service-to-exports ratio (in percent)	**15.2**	**15.4**	**15.3**			**20.7**	**24.1**	**26.3**	**26.8**	**23.8**	**24.9**		**43.9**	**20.0**
PPG debt service-to-revenue ratio (in percent)	**5.1**	**6.0**	**8.0**			**11.2**	**12.2**	**12.4**	**11.9**	**11.1**	**11.0**		**13.9**	**4.2**
Total gross financing need (Billions of U.S. dollars)	0.0	0.1	0.0			0.6	0.5	0.4	0.3	0.3	0.3		0.7	0.9
Non-interest current account deficit that stabilizes debt ratio	3.7	-4.0	-2.9			0.7	4.5	5.4	5.1	4.9	4.8		4.0	2.3
Key macroeconomic assumptions														
Real GDP growth (in percent)	6.7	6.0	6.9	6.8	1.2	4.5	5.0	5.5	6.0	6.0	6.0	5.5	6.0	6.0
GDP deflator in US dollar terms (change in percent)	1.8	-19.8	-17.6	3.5	15.3	0.0	-5.2	2.5	2.3	2.3	2.2	0.7	1.9	1.9
Effective interest rate (percent) 5/	3.2	4.8	2.2	3.4	1.2	2.3	2.3	4.1	4.5	4.5	4.2	3.6	3.5	5.6
Growth of exports of G&S (US dollar terms, in percent)	-10.7	-1.4	12.0	-3.4	21.4	0.5	0.4	2.8	3.1	14.2	3.1	4.0	3.7	4.7
Growth of imports of G&S (US dollar terms, in percent)	-19.1	-19.9	-10.4	5.1	25.7	12.4	8.3	10.0	7.0	7.7	5.0	8.4	6.9	5.5
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	0.1	2.0	-5.7	23.4	23.6	24.4	11.3	34.8	39.8
Government revenues (excluding grants, in percent of GDP)	26.9	27.0	25.4			23.7	25.7	26.1	26.3	26.4	26.5		29.8	30.4
Aid flows (in Billions of US dollars) 7/	0.1	0.2	0.2			0.4	0.3	0.3	0.4	0.4	0.5		0.9	0.8
of which: Grants	0.1	0.2	0.2			0.3	0.3	0.3	0.3	0.3	0.4		0.4	0.3
of which: Concessional loans	0.0	0.0	0.0			0.1	0.0	0.0	0.1	0.1	0.1		0.5	0.5
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.5	3.7	3.0	4.4	4.4	4.5		4.6	2.0
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			34.3	39.3	32.5	61.7	63.7	65.2		57.8	58.1
Memorandum items:														
Nominal GDP (Billions of US dollars)	9.2	7.9	6.9			7.2	7.2	7.8	8.4	9.2	9.9		14.6	31.7
Nominal dollar GDP growth	8.7	-15.0	-11.9			4.5	-0.4	8.2	8.4	8.4	8.3	6.2	8.0	8.0
PV of PPG external debt (in Billions of US dollars)			1.8			2.2	2.5	2.8	2.9	3.1	3.2		3.5	5.6
(PVt-PVt-1)/GDPt-1 (in percent)						5.7	4.0	5.1	1.4	1.3	1.1	3.1	0.4	1.0
Gross workers' remittances (Billions of US dollars)	3.4	2.3	1.9			2.2	2.6	2.9	3.2	3.4	3.6		4.8	8.5
PV of PPG external debt (in percent of GDP + remittances)	...	...	20.3			25.4	25.7	26.9	25.7	24.7	23.7	25.4	18.3	14.1
PV of PPG external debt (in percent of exports + remittances)	...	...	64.0			77.0	72.0	74.0	71.1	68.4	67.2		57.4	53.6
Debt service of PPG external debt (in percent of exports + remittances)	...	...	5.1			6.2	6.5	6.5	6.3	5.9	6.0		9.8	3.9

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate; and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 6. Tajikistan: Public Sector Debt Sustainability Framework, Strong Reform Scenario, 2014–37

	Actual					Estimate						Projections		
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037
Public sector debt 1/	27.9	34.7	42.3			52.4	54.6	53.1	49.6	46.4	44.0		35.8	26.3
of which: foreign-currency denominated	24.4	31.5	32.9			39.7	40.6	41.7	40.4	39.1	37.7		33.1	25.9
Change in public sector debt	-1.1	6.8	7.6			10.2	2.2	-1.5	-3.5	-3.2	-2.4		-1.3	-0.6
Identified debt-creating flows	-1.0	7.6	10.2			11.0	4.4	-3.2	-3.4	-3.2	-3.1		-0.8	-2.3
Primary deficit	-0.4	1.3	10.0	2.9	3.3	5.4	3.6	-0.3	-0.7	-0.8	-0.9	1.1	1.2	-0.6
Revenue and grants	28.4	29.9	27.9			27.2	29.2	29.4	29.8	30.0	30.2		32.6	31.4
of which: grants	1.5	3.0	2.5			3.5	3.5	3.4	3.5	3.6	3.7		2.8	1.0
Primary (noninterest) expenditure	27.9	31.3	37.9			32.7	32.8	29.1	29.1	29.2	29.3		33.8	30.8
Automatic debt dynamics	-0.2	6.3	0.3			1.5	-2.7	-2.9	-2.6	-2.4	-2.2		-2.0	-1.7
Contribution from interest rate/growth differential	-1.9	-1.3	-2.2			-2.2	-2.5	-2.7	-2.6	-2.3	-2.1		-2.0	-1.7
of which: contribution from average real interest rate	-0.1	0.2	0.0			-0.3	0.0	0.2	0.4	0.5	0.5		0.1	-0.1
of which: contribution from real GDP growth	-1.8	-1.6	-2.2			-1.8	-2.5	-2.8	-3.0	-2.8	-2.6		-2.1	-1.5
Contribution from real exchange rate depreciation	1.7	7.6	2.5			3.7	-0.3	-0.2	0.0	-0.1	-0.1		...	...
Other identified debt-creating flows	-0.3	0.0	0.0			4.0	3.5	0.0	0.0	0.0	0.0		0.0	0.0
Privatization receipts (negative)	-0.3	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			4.0	3.5	0.0	0.0	0.0	0.0		0.0	0.0
Residual, including asset changes	-0.1	-0.8	-2.7			-0.8	-2.3	1.7	-0.1	0.1	0.7		-0.5	1.6
Other Sustainability Indicators														
PV of public sector debt	...	...	35.2			45.8	48.8	48.4	44.7	41.3	38.7	44.6	27.0	18.2
of which: foreign-currency denominated	...	...	25.8			33.1	34.8	37.0	35.4	33.9	32.4		24.3	17.9
of which: external	...	...	25.8			33.1	34.8	37.0	35.4	33.9	32.4		24.3	17.9
PV of contingent liabilities (not included in public sector de	...	...	...			...	...	...	...	...	...		...	...
Gross financing need 2/	1.7	3.9	12.4			9.2	8.4	4.6	3.9	3.5	2.6		5.6	0.7
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	126.2			168.4	167.2	164.4	149.7	137.5	128.1		82.8	58.0
PV of public sector debt-to-revenue ratio (in percent)	...	...	138.4			193.1	190.3	185.7	169.8	156.5	146.1		90.5	60.0
of which: external 3/	...	...	101.5			139.5	135.8	141.9	134.7	128.6	122.4		81.6	58.9
Debt service-to-revenue and grants ratio (in percent) 4/	7.3	8.2	8.8			13.3	16.2	16.8	15.6	14.3	11.6		13.6	4.3
Debt service-to-revenue ratio (in percent) 4/	7.8	9.1	9.6			15.2	18.4	19.0	17.7	16.3	13.2		14.9	4.4
Primary deficit that stabilizes the debt-to-GDP ratio	0.6	-5.5	2.4			-4.7	1.5	1.2	2.8	2.3	1.5		2.5	0.0
Key macroeconomic and fiscal assumptions														
Real GDP growth (in percent)	6.7	6.0	6.9	6.8	1.2	4.5	5.0	5.5	6.0	6.0	6.0	5.5	6.0	6.0
Average nominal interest rate on forex debt (in percent)	1.5	1.5	1.5	1.5	0.1	2.2	2.9	3.5	3.8	3.8	3.8	3.3	2.5	1.4
Average real interest rate on domestic debt (in percent)	-1.5	4.3	-1.6	-1.3	3.2	-5.0	-2.8	-2.9	-2.8	-2.6	-2.4	-3.1	-2.0	0.5
Real exchange rate depreciation (in percent, + indicates depreciation)	7.3	32.9	8.4	-0.1	16.1	11.7	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	5.5	0.1	5.3	12.1	9.4	10.0	6.0	6.0	6.0	6.0	6.0	6.7	6.0	6.0
Growth of real primary spending (deflated by GDP deflator, in percent)	9.3	18.6	29.4	5.8	10.3	-9.9	5.6	-6.3	5.8	6.4	6.4	1.3	9.6	5.8
Grant element of new external borrowing (in percent)	...	...	...	...	...	0.1	2.0	-5.7	23.4	23.6	24.4	11.3	34.8	39.8

Sources: Country authorities; and staff estimates and projections.
1/ Coverage of public sector is general government and gross debt is used.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 7. Tajikistan: Sensitivity Analysis for Key Indicators of Public Debt—Strong Reform, 2017–37

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	46	49	48	45	41	39	27	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	46	47	49	48	47	46	37	45
A2. Primary balance is unchanged from 2017	46	50	54	55	56	57	55	76
A3. Permanently lower GDP growth 1/	46	49	49	45	42	40	30	28
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	46	48	48	44	41	38	26	16
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	46	51	55	51	48	45	32	21
B3. Combination of B1-B2 using one half standard deviation shocks	46	49	52	48	44	41	27	15
B4. One-time 30 percent real depreciation in 2018	46	63	62	58	54	51	35	26
B5. 10 percent of GDP increase in other debt-creating flows in 2018	46	55	55	51	48	45	33	22
PV of Debt-to-Revenue Ratio 2/								
Baseline	168	167	164	150	138	128	83	58
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	168	163	167	160	156	154	113	143
A2. Primary balance is unchanged from 2017	168	172	183	183	185	189	169	241
A3. Permanently lower GDP growth 1/	168	168	166	152	140	132	92	90
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	168	166	163	148	135	125	79	51
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	168	174	187	172	158	148	98	67
B3. Combination of B1-B2 using one half standard deviation shocks	168	168	176	160	146	136	83	47
B4. One-time 30 percent real depreciation in 2018	168	217	212	194	179	168	109	83
B5. 10 percent of GDP increase in other debt-creating flows in 2018	168	190	187	172	159	150	101	70
Debt Service-to-Revenue Ratio 2/								
Baseline	13	16	17	16	14	12	14	4
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	13	16	16	15	14	12	14	9
A2. Primary balance is unchanged from 2017	13	16	17	16	16	13	17	15
A3. Permanently lower GDP growth 1/	13	16	17	16	14	12	14	6
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	13	16	17	15	14	11	13	4
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	13	16	17	16	15	12	15	5
B3. Combination of B1-B2 using one half standard deviation shocks	13	16	17	16	15	12	14	4
B4. One-time 30 percent real depreciation in 2018	13	19	22	20	19	16	20	9
B5. 10 percent of GDP increase in other debt-creating flows in 2018	13	16	18	17	15	12	16	6

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

Table 8. Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt—Strong Reform, 2017–37

(In percent)

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of debt-to-GDP+remittances ratio								
Baseline	25	26	27	26	25	24	18	14
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	25	24	26	26	26	26	22	15
A2. New public sector loans on less favorable terms in 2017-2037 2/	25	26	28	28	27	27	25	23
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	25	25	26	25	24	23	18	14
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	25	27	31	30	29	28	21	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	25	27	31	29	28	27	21	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	25	35	43	36	34	33	25	15
B5. Combination of B1-B4 using one-half standard deviation shocks	25	25	30	28	27	26	20	14
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	25	33	34	33	31	30	23	18
PV of debt-to-exports+remittances ratio								
Baseline	77	72	74	71	68	67	57	54
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	77	70	75	76	77	80	81	80
A2. New public sector loans on less favorable terms in 2017-2037 2/	77	74	78	76	75	75	78	86
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	77	71	73	70	67	66	56	52
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	77	81	97	92	89	88	73	60
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	77	71	73	70	67	66	56	52
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	77	134	165	99	95	94	78	58
B5. Combination of B1-B4 using one-half standard deviation shocks	77	72	84	80	78	76	64	55
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	77	71	73	70	67	66	56	52
PV of debt-to-revenue ratio								
Baseline	139	136	142	135	129	122	82	59
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	139	123	131	130	128	127	92	59
A2. New public sector loans on less favorable terms in 2017-2037 2/	139	139	149	144	141	137	111	95
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	139	133	138	131	125	119	79	57
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	139	143	165	156	149	142	93	60
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	139	144	174	165	157	150	99	71
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	139	163	197	187	179	171	111	64
B5. Combination of B1-B4 using one-half standard deviation shocks	139	134	161	153	145	138	92	61
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	139	195	204	193	184	175	116	83

Table 8. Tajikistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt—Strong Reform, 2017–37 (concluded)

(In percent)

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
Debt service-to-exports+remittances ratio								
Baseline	6	6	6	6	6	6	10	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	6	6	6	6	6	6	10	6
A2. New public sector loans on less favorable terms in 2017-2037 2/	6	6	5	6	6	6	7	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	6	6	6	6	6	6	10	4
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	6	7	7	7	7	7	12	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	6	6	6	6	6	6	10	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	6	10	11	7	7	7	12	5
B5. Combination of B1-B4 using one-half standard deviation shocks	6	6	7	7	6	7	11	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	6	6	6	6	6	6	10	4
Debt service-to-revenue ratio								
Baseline	11	12	12	12	11	11	14	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	11	11	11	10	10	10	12	4
A2. New public sector loans on less favorable terms in 2017-2037 2/	11	12	11	11	10	10	11	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	11	12	12	12	11	11	14	4
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	11	12	13	13	12	12	15	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	11	13	15	15	14	14	17	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	11	12	13	14	13	13	17	6
B5. Combination of B1-B4 using one-half standard deviation shocks	11	12	13	13	12	12	15	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	11	18	18	17	16	16	20	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	24	24	24	24	24	24	24	24

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.